ME380 INDIVIDUAL STUDENT PORTFOLIO

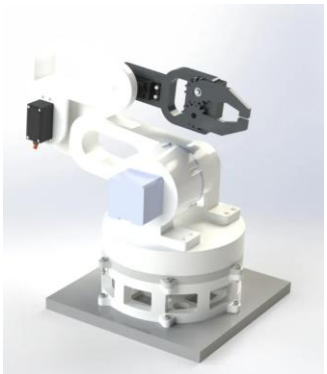
DESIGN OBJECTIVES:

The objectives for the development of this project were:

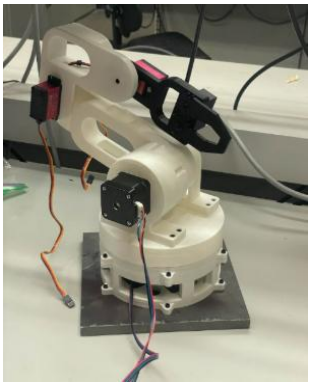
- To create a robot arm capable of lifting, transporting, and place a minimum payload of 1:3 the weight of the entire robot arm.
- Complete the required Benchy pick-and-place cycle in < 60 seconds.
- Achieve a repeatable positional accuracy of < 1 cm.
- Use only one approved power supply unit, drawing < 24 V.

These objectives outlined and guided the development of:

OUR DESIGN:



A



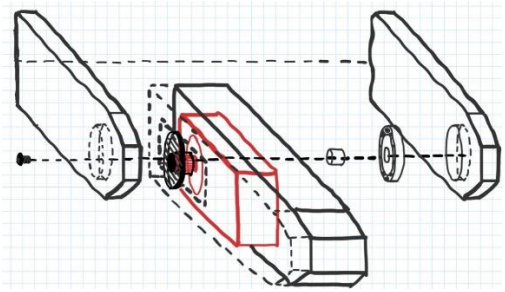
B

C

Figure 1: A - Final Rendered Assembly, B - Final Un-wired Assembly, C - Final Wired Assembly

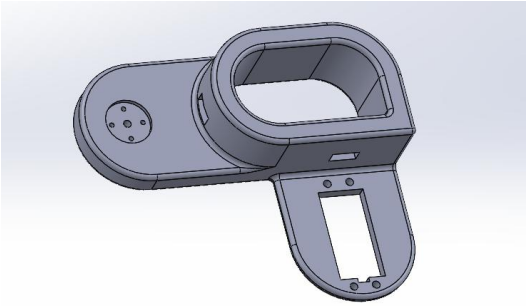
MAIN INDIVIUAL CONTRIBUTIONS:

Preliminary Model and Joint Design



- Designed joint connections to link the arm segments together at servo joints in drawing form, then in CAD.

Elbow and Joint Connection Design



- Designed the elbow arm segment and helped design the shoulder arm segment in CAD.

Miniature Control Unit

Base Plate Manufacturing

- Fully designed a miniature control arm in CAD, controlled by potentiometers, to mimic the movements of the arm as a potential method of control.

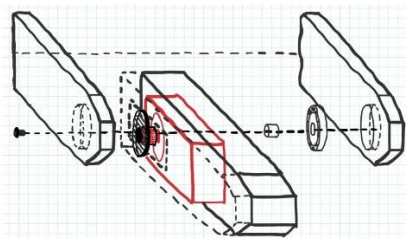
- Manufactured the holes to mount the base to on a mill and tapped them to the correct size in only 1 attempt.

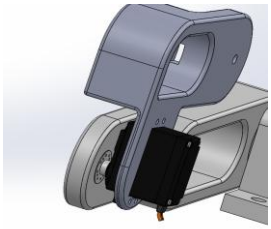
IMPACT ON PROJECT OUTCOME:

- **Enabled Successful and Smooth Project Integration:**
  - Designing the servo connections made them a standard for all the arms, even the pieces I did not work on (like the wrist). Creating this standard allowed for all the connections to be the same, as my initial design could be referred to, allowing for smooth integration.
- **Optimized For Weight and Minimized Complexity to Meet Design Constraints:**
  - Overall, my design choices decreased mass, in both the joints and arms themselves to help meet our strength to weight ratio requirements, and made the design as simple as possible, while allowing for some extra design features that improved wire keeping.

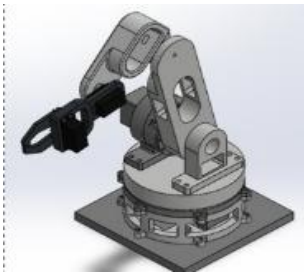
TECHNICAL SUMMARY OF DESIGN PROCESS:

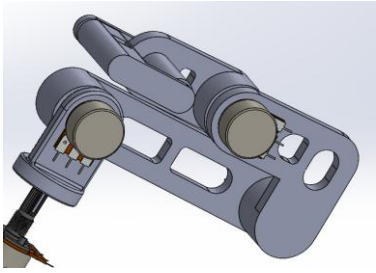
1) **Preliminary Model and Joint Design**
   - **a.** Defined the joint connection between servos arm segments in initial sketches and established clear rotational axes and mounting points.
   - **b.** These sketches guided the development of the CAD, and the standardized joints to implement across all servo connections.
   - **c.** Established consistent servo mounting template in CAD, then implemented them in each joint.
2) **Elbow and Joint Connection Design**
   - **a.** Developed elbow solely and helped with shoulder segment design through an iterative CAD design process.
   - **b.** Motor orientations were considered to allow for the full 180 degrees of motion for the servos.
   - **c.** Each iteration was checked for assembly and feasibilities regarding stresses on the joint until a final design.





3) **Miniature Control Unit**
   - **a.** Designed a scaled down 3D modelled a miniature control arm to control the arm using potentiometers.
   - **b.** After the pieces were 3D printed, the parts were assembled to ensure the correct tolerances were met and the arm behaves as intended.
   - **c.** The potentiometers were mounted and then the wiring was done by soldering. Unfortunately, this control was not used due to control limitations.





4) **Base-Plate Manufacturing**
   - **a.** Worked with the team to come up with a solution for counterweight, and a base plate was the solution that was come up with.
   - **b.** The dimensions were created by a great team member, and I manufactured the holes using a mill and automatic tapper, tapped to m3, ensuring maximum accuracy.

LEARNING OUTCOMES:

1) Improved proficiency in CAD modelling and GD&T, especially in multipart assemblies like the miniature arm, and modeling tolerances for clearance, slide, and interference for shafts.
2) Gained a better understanding of integrating mechanical, control, and electrical into a single design, keeping into account wiring, space, and the limitations of the electrical and control parts while mechanically design parts.

SUBMISSION:

We verify the accuracy of the individual contributions to the ME380 design project and, unless otherwise stated and properly referenced, the content of this document represents original engineering design work completed by the author and project group.

Author: _____ Verified By: _____